

Shubham Sharma

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TECHNICAL SKILLS

Languages & Frameworks: Python, Java, TensorFlow, Keras, PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib, OpenCV

AI / ML & GenAI: LLMs, RAG, LLM Fine-Tuning (QLoRA / LoRA), Prompt Engineering, Agentic Systems, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, HuggingFace Transformers, vLLM, Ollama, MCP

ML Lifecycle & MLOps: Data Preprocessing, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Experiment Tracking (MLflow / W&B), Model Deployment, CI/CD

Vector Databases & APIs: Pinecone, FAISS, FastAPI, Flask, REST APIs

Cloud, DevOps & Tools: AWS (SageMaker, EC2, Lambda, SQS, DynamoDB, ALB, ASG, S3), Oracle Cloud (OCI), Docker, React.js, Next.js, Git, Bash

EXPERIENCE

Google Developer Student Clubs (GDSC)

Sept 2025 – Present

Machine Learning Lead

Bhopal, India

- Spearheaded **5+ end-to-end ML projects** spanning Multi-Agent Systems, RAG Agents, and Agentic AI Workflows, driving a **30% increase** in active community engagement across the chapter.
- Designed and facilitated technical workshops on **LangChain, LangGraph, and LLM deployment**, mentoring **50+ students** in building and shipping production-ready AI pipelines.
- Established a structured ML curriculum covering RAG architectures and automation workflows, measurably elevating the chapter's technical output and university-wide visibility.

Entrepreneurship Cell (E-Cell)

Sept 2024 – Present

Tech Team Member

Bhopal, India

- Engineered end-to-end automation workflows for **social media, email pipelines, and event management**, boosting operational efficiency by **40%** and eliminating manual overhead.
- Architected and launched the official club website using **React.js and Next.js**, growing monthly user engagement by over **200%** through improved UX and automated content delivery.

PROJECTS

QLoRA Customer Support LLM | Llama 3.1 8B, SageMaker, QLoRA, vLLM, MLflow

🐙 /customer_support_agent

- Fine-tuned **Llama 3.1 8B Instruct** with **QLoRA** (4-bit NF4) on AWS SageMaker, training only **0.75%** of parameters to reach **83.2% eval token accuracy** in 4.4 hours under 20 GB VRAM, with all runs tracked in **MLflow**.
- Deployed the model via **vLLM + FastAPI** on EC2 behind an Application Load Balancer with Auto Scaling, and built a real-time escalation pipeline (**SQS** → **Lambda** → **DynamoDB**) routing complaints and refunds to a human queue.

Local Multi-Agent Dev System | Python, LangGraph, Ollama, SLMs

🐙 /Local_multi_agent_system

- Engineered a fully **offline 7-agent software-development pipeline** on **LangGraph**, orchestrating small language models (**≤9B**) via **Ollama** to turn natural-language goals into runnable code, tests, migrations, and docs.
- Designed an inspectable filesystem-based memory layer plus an automated **exec-debug** loop with surgical patching, retry budgets, and error caps for self-healing task execution.

LawApp — Legal Aid RAG Agent | LangChain, LangGraph, Pinecone, Gemini, Streamlit

🐙 /LawApp

- Architected a **RAG agent** grounded in the Indian Constitution and Bharatiya Nyaya Sanhita, combining **Gemini 2.5 Flash**, **Pinecone** retrieval, and BGE embeddings, with a context-only system prompt that refuses to hallucinate.
- Built a stateful **LangGraph** tool-using agent with **MemorySaver** conversational memory, containerized with Docker and deployed publicly on Hugging Face Spaces.

Diabetic Foot Ulcer Detection | TensorFlow, Keras, ResNet50, OpenCV

🐙 /Diabetic_foot_ulcer

- Developed a **CNN classifier** via **ResNet50 transfer learning** on thermographic foot images (Kaggle), achieving **93.3% validation accuracy**, **95.9% AUC**, and **97.7% precision** for non-invasive early ulcer screening.
- Applied data augmentation, batch normalization, and dropout regularization to improve robustness; published the full reproducible pipeline on GitHub for open medical-imaging research.

EDUCATION

VIT Bhopal University

2023 – 2027

B.Tech, Computer Science — Specialization: Artificial Intelligence & Machine Learning

Bhopal, Madhya Pradesh

CERTIFICATIONS

Oracle Cloud Infrastructure Certified Generative AI Professional

Oracle

Deep Learning Specialization

DeepLearning.AI

Machine Learning Specialization

Stanford Online & DeepLearning.AI

Java Certified Foundations Associate

Oracle